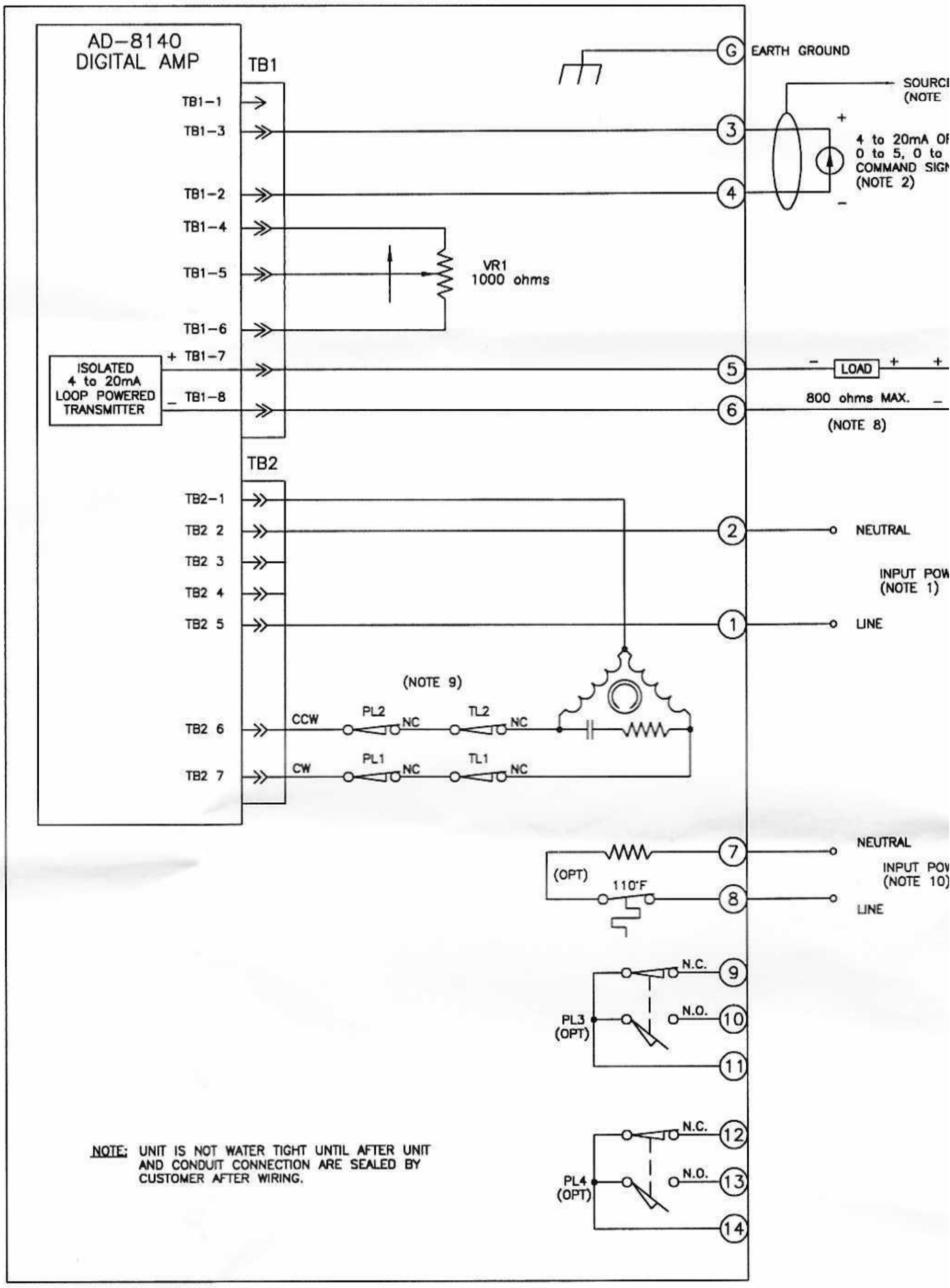
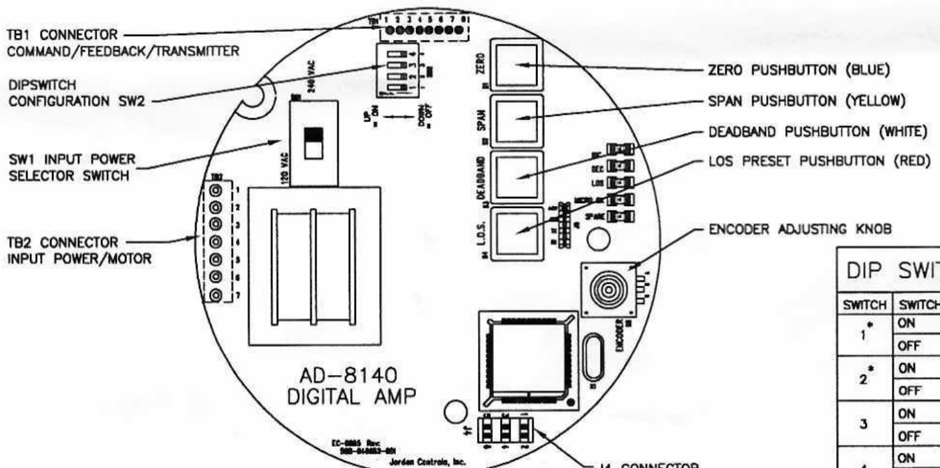
SM-1700 / AD-8140



NOTE: UNIT IS NOT WATER TIGHT UNTIL AFTER UNIT AND CONDUIT CONNECTION ARE SEALED BY CUSTOMER AFTER WIRING.



D

J4-5 AND J4-6 SHOULD BE JUMPER
WHEN VOLTAGE COMMAND INPUT IS SE
NOT REQUIRED FOR CURRENT COMMAN
MAY BE PARKED BETWEEN J4-3 AND

DIP SWITCH CONFIGURATIONS

SWITCH	SWITCH POS	FUNCTION
1*	ON	CURRENT COMMAND
	OFF	0-5V/0-10V VOLTAGE COMMAN
2*	ON	0-5V VOLTAGE COMMAND
	OFF	CURRENT COMMAND/0-10V CO
3	ON	LOS LOCK-IN-PLACE
	OFF	LOS GO TO PREST POSITION
4	ON	DYNAMIC BRAKE ON
	OFF	DYNAMIC BRAKE OFF

ON

A

NOTES: REF: ALL ROTATION NOTES ARE VIEWED FACING OUTPUT SHAFT.

WARNING! BEFORE APPLYING INPUT POWER, CHECK FOR PROPER SELECTION OF SLIDE SWITCH SW1. OPERATING VOLTAGE CHANGES CANNOT BE MADE SIMPLY BY CHANGING THE POSITION OF THE AMPLIFIER VOLTAGE SWITCH. MOTOR AND CAPACITOR MUST BE CHANGED AS WELL. REFER TO ACTUATOR NAME PLATE FOR PROPER VOLTAGE.

1) INPUT POWER: FOR 120VAC INPUT MOVE SLIDE SWITCH (SW1) TO THE 120VAC POSITION. (SEE WARNING) FOR 240VAC OR 220VAC INPUT MOVE SLIDE SWITCH (SW1) TO THE 240VAC POSITION. (SEE WARNING) FOR INPUT POWER REFER TO ACTUATOR NAMEPLATE.

2) COMMAND INPUT: CURRENT COMMAND:
FOR 4 to 20mA COMMAND INPUT SIGNAL, SELECT DIP SWITCH SW2–1 IN THE ON POSITION.

VOLTAGE COMMAND:
FOR 0–10 VDC COMMAND SIGNAL INPUT, SELECT DIP SWITCH SW2–1 IN THE OFF POSITION AND SELECT DIP SWITCH SW2–2 IN THE OFF POSITION.

FOR 0–5 VDC COMMAND SIGNAL INPUT, SELECT DIP SWITCH SW2–1 IN THE OFF POSITION AND SELECT DIP SWITCH SW2–2 IN THE ON POSITION.

3) SHIELDING: SHIELDED CABLE IS REQUIRED FOR ALL COMMAND AND FEEDBACK SIGNAL WIRING. TERMINATE GROUND AT SOURCE COMMON.

4) COMMAND CALIBRATION APPLY LOW COMMAND TO UNIT, NORMALLY 4mA. DEPRESS S1 (BLUE) AND S4 (RED) UNTIL SPARE LED ILLUMINATES.
APPLY HIGH COMMAND TO UNIT, NORMALLY 20mA. DEPRESS S2 (YELLOW) AND S4 (RED) UNTIL SPARE LED ILLUMINATES.

B

5) ENDPOINT CALIBRATION: AN INCREASING COMMAND SIGNAL WILL RESULT IN "CW" ROTATION OF THE OUTPUT SHAFT. SET COMMAND SIGNAL TO A MINIMUM (4mA) AND ADJUST ZERO TO THE DESIRED POSITION BY HOLDING DOWN THE S1 SWITCH (BLUE) AND ROTATING THE ADJUSTING KNOB UNTIL AMPLIFIER NULLS JUST BEFORE THE "CCW" POSITION LIMIT "TRIPS". SET COMMAND SIGNAL TO MAXIMUM (20mA) AND ADJUST SPAN TO THE DESIRED POSITION BY HOLDING DOWN THE S2 SWITCH (YELLOW) AND ROTATING THE ADJUSTING KNOB CW UNTIL AMPLIFIER NULLS JUST BEFORE THE "CW" POSITION LIMIT "TRIPS". IT MAY BE NECESSARY TO REPEAT THESE STEPS UNTIL PROPER ACCURACY IS ACHIEVED.

REVERSE ACTING: FOR AN INCREASING COMMAND RESULTING IN "CCW" ROTATION SET COMMAND SIGNAL TO A MINIMUM (4mA) AND ADJUST ZERO TO THE DESIRED POSITION BY HOLDING DOWN THE S1 SWITCH (BLUE) AND ROTATING THE ADJUSTING KNOB UNTIL AMPLIFIER NULLS JUST BEFORE THE "CW" POSITION LIMIT "TRIPS". SET COMMAND SIGNAL TO MAXIMUM (20mA) AND ADJUST SPAN TO THE DESIRED POSITION BY HOLDING DOWN THE S2 SWITCH (YELLOW) AND ROTATING THE ADJUSTING KNOB UNTIL AMPLIFIER NULLS JUST BEFORE THE "CCW" POSITION LIMIT "TRIPS". IT MAY BE NECESSARY TO REPEAT THESE STEPS UNTIL PROPER ACCURACY IS ACHIEVED.

6) LOSS OF SIGNAL: IF LOSS OF SIGNAL GO TO PRESET IS SELECTED (SW2–3 IN THE OFF POSITION), ADJUST PRESET BY HOLDING DOWN THE LOS PUSHBUTTON S4 (RED) AND ROTATING THE ADJUSTING KNOB TO THE DESIRED POSITION.

C

7) DEADBAND: DEADBAND ADJUSTS THE "SENSITIVITY" OF THE SERVO LOOP. DEPRESS AND HOLD S3 SWITCH (WHITE) AND TURNING THE ADJUSTING KNOB CW INCREASES DEADBAND. DEPRESS AND HOLD S3 (WHITE) AND TURNING THE ADJUSTING KNOB CCW DECREASES DEADBAND.

8) TRANSMITTER: INPUT POWER TO TERMINAL 5 POSITIVE (+) AND 6 NEGATIVE (–) REQUIRES AN EXTERNAL REGULATED POWER SUPPLY IN THE RANGE OF 12.0 (MIN.) TO 36VDC (MAX.) AND A LOAD (CUSTOMER SUPPLIED) CONNECTED IN SERIES WITH THE POWER SUPPLY AS SHOWN. TRANSMITTER SIGNAL WILL FOLLOW THE COMMAND SIGNAL. (EX: INC COMMAND = INC FEEDBACK)

$$\frac{\text{POWER SUPPLY VOLTAGE} - \text{8VDC}}{\text{0.020A MAX.}} = \text{LOAD RESISTANCE.}$$

$$(\,0.020\text{A}\,)^2 \text{X LOAD RESISTANCE} = \text{MINIMUM WATT RATING OF RESISTOR}$$

TRANSMITTER WILL PRODUCE AN INCREASING SIGNAL AS ACTUATOR MOVES TOWARD SPAN, AND DECREASING SIGNAL AS ACTUATOR MOVES TOWARD ZERO.

APPLY LOW COMMAND TO UNIT, NORMALLY 4mA. DEPRESS S1 (BLUE) AND S4 (RED) UNTIL SPARE LED ILLUMINATES. WHILE HOLDING SWITCHES ROTATE ADJUSTING KNOB CW TO INCREASE 4mA POINT OR CCW TO DECREASE 4mA POINT. APPLY HIGH COMMAND TO UNIT, NORMALLY 20mA. DEPRESS S2 (YELLOW) AND S4 (RED) UNTIL SPARE LED ILLUMINATES. WHILE HOLDING SWITCHES ROTATE ADJUSTING KNOB CW TO INCREASE 20mA POINT OR CCW TO DECREASE 20mA POINT.

D

9) POSITION LIMITS: ALL POSITION LIMITS SHOWN AT MID-TRAVEL OF THE ACTUATOR.

10) HEATER: INPUT POWER IS REQUIRED TO THE HEATER FOR CONDENSATION PROTECTION OF THE ACTUATOR. REF. ACTUATOR NAMEPLATE FOR HEATER INPUT VOLTAGE.

11) GROUND: GROUND IS IDENTIFIED BY A GREEN SCREW ON HOUSING OF ACTUATOR. GROUND IS REQUIRED.

12) INFORMATION: REFER TO INSTRUCTION MANUAL FOR SET-UP INSTRUCTIONS.

ER.
JUMPER
BUT

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REV	DESCRIPTION	DATE
D	ECR16959 ADDED JUMPER INFO FOR J4	JPS 05/23/05
C	ECR–16426 RENAMED TO AD-8140	E.NOWAK 12/23/03
B	ECR 16101	DAH 10/17/02
A	INITIAL RELEASE	E.NOWAK 6/19/02

TOLERANCES
UNLESS OTHERWISE SPECIFIED
.XX ±.02 .XXX ±.005
ANGULAR ± 1°

SCALE– NONE		
DRAWN– E.N.	PROD DEV–	PROD ENG– B.Z.
DATE– 6/19/02	DATE–	DATE– 6/19/02

DO NOT SCALE

Jordan Controls Inc.

MILWAUKEE, WISCONSIN, USA

TITLE– WIRING DIAGRAM, INTERCONNECT SM-1700 / AD-8140 DIGITAL

FOR– POT FEEDBACK STANDARD PL'S

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